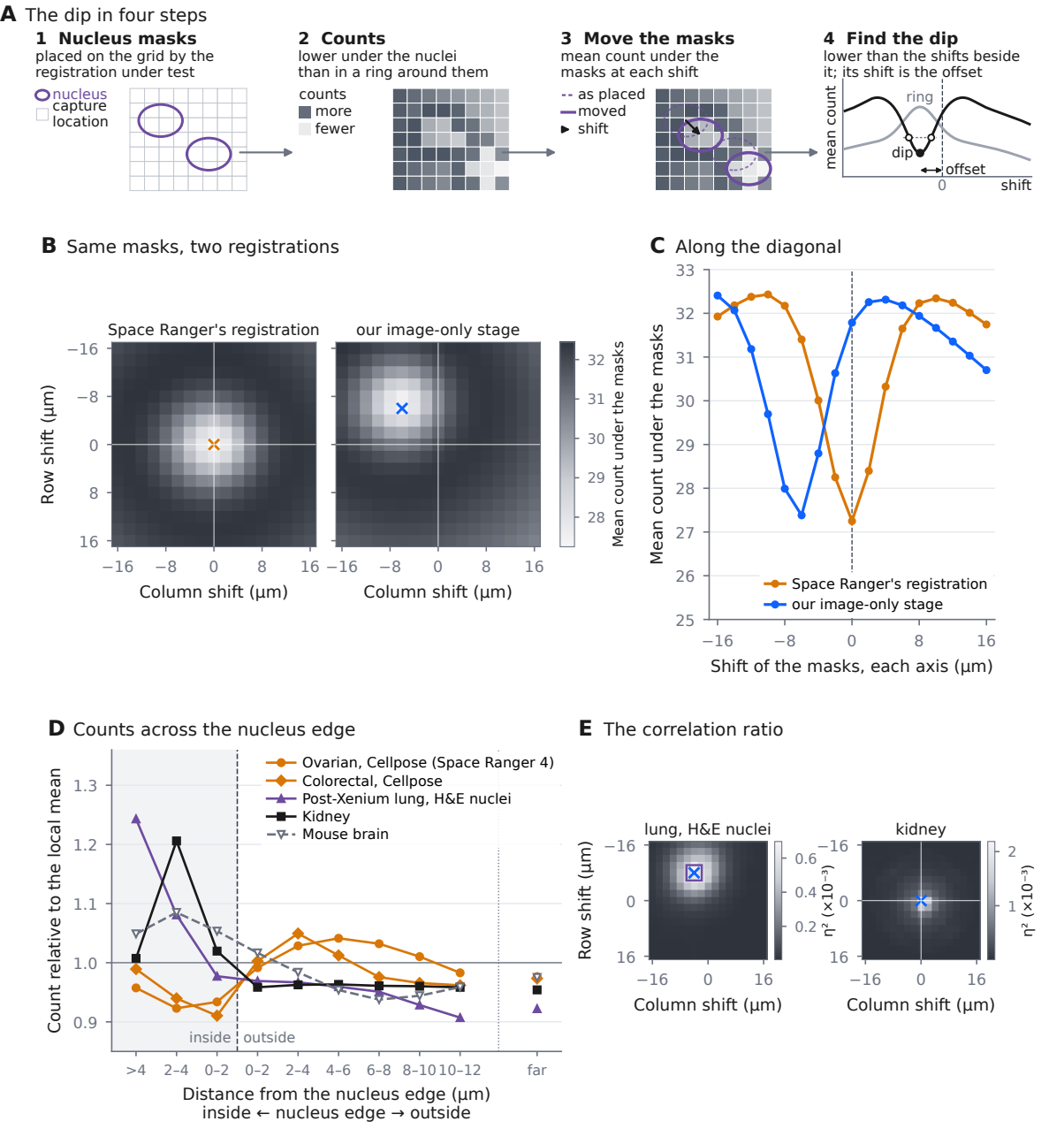

A The dip in four steps
1 Nucleus masks
placed on the grid by the registration under test
nucleus
capture location
2 Counts
lower under the nuclei than in a ring around them
counts
more
fewer
3 Move the masks
mean count under the masks at each shift
as placed
moved
shift
4 Find the dip
lower than the shifts beside it; its shift is the offset
mean count
ring
dip
offset
0
shift
B Same masks, two registrations
Space Ranger's registration
our image-only stage
Row shift (µm)
Column shift (µm)
Mean count under the masks
C Along the diagonal
Mean count under the masks
Space Ranger's registration
our image-only stage
Shift of the masks, each axis (µm)
D Counts across the nucleus edge
Ovarian, Cellpose (Space Ranger 4)
Colorectal, Cellpose
Post-Xenium lung, H&E nuclei
Kidney
Mouse brain
Count relative to the local mean
inside
outside
Distance from the nucleus edge (µm)
inside ← nucleus edge → outside
E The correlation ratio
lung, H&E nuclei
kidney
Row shift (µm)
Column shift (µm)
η² (×10⁻³)